

FUKUE, Japan

WMO: 478430

Lat: 32.6947N

Lon: 128.8269E

Elev: 27

StdP: 101.00

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	2.1	-6.4	2.2	4.1	-5.3	2.4	4.7	9.5	6.3	8.7	6.8	3.2	210	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	31.8	26.0	31.0	25.9	30.2	25.7	27.1	29.9	26.7	29.4	26.3	29.0	3.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.4	21.9	28.7	26.0	21.4	28.4	25.5	20.8	28.2	85.8	30.2	83.9	29.5	82.2	29.0	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.9	7.0	-1.1	33.5	1.0	1.0	-1.9	34.2	-2.5	34.9	-3.1	35.4	-3.8	36.2		
(5)			-2.4	27.5	0.9	0.7	-3.1	28.0	-3.6	28.4	-4.1	28.8	-4.8	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	7.4	8.1	10.7	14.8	18.9	22.2	26.5	27.6	24.3	19.8	14.5	9.6		
	DBStd	7.33	2.72	3.03	2.87	2.63	1.91	1.87	1.95	1.53	2.11	2.42	3.05	2.92		
	HDD10.0	229	91	68	26	1	0	0	0	0	0	0	2	43		
	HDD18.3	1392	341	286	238	109	17	0	0	0	0	13	119	270		
	CDD10.0	2811	8	16	46	144	274	365	512	544	430	303	137	31		
	CDD18.3	932	0	0	0	3	33	115	254	286	180	57	4	0		
	CDH23.3	7373	0	0	0	1	74	446	2400	3074	1206	168	3	0		
	CDH26.7	1990	0	0	0	0	2	22	654	1062	242	8	0	0		
(14) Wind	WSAvg	3.1	3.5	3.5	3.6	3.2	2.8	2.7	3.1	2.9	2.9	3.0	3.0	3.3		
(15) Precipitation	PrecAvg	2277	97	111	166	218	217	328	281	232	257	123	140	107		
	PrecMax	3037	230	224	400	407	390	651	500	506	613	325	344	226		
	PrecMin	1565	13	19	58	96	92	54	30	61	69	3	5	15		
	PrecStd	366	51	52	68	82	87	140	128	102	142	92	85	58		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	18.7	20.0	23.0	26.6	28.4	32.5	33.3	31.1	27.7	23.7	19.3		
		MCWB	15.3	16.4	16.1	17.2	18.2	23.2	26.1	26.3	25.2	23.2	19.3	16.4		
	2%	DB	15.4	17.1	18.6	21.7	25.1	27.2	31.4	32.3	30.1	26.2	22.2	17.6		
		MCWB	12.8	14.8	15.1	16.9	18.1	22.8	26.1	26.1	25.0	21.8	18.3	14.2		
	5%	DB	13.7	15.5	17.6	20.6	24.0	26.5	30.6	31.5	29.1	25.1	21.0	16.2		
		MCWB	10.5	12.5	14.5	16.6	18.2	22.6	25.9	26.0	24.4	20.5	17.4	12.7		
	10%	DB	12.1	13.6	16.3	19.7	23.0	25.7	29.8	30.8	28.1	24.1	19.8	14.9		
		MCWB	8.9	10.4	13.1	15.9	18.2	22.4	25.7	25.8	23.8	19.5	16.2	11.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.9	17.0	17.6	20.4	22.7	25.6	27.7	27.7	26.7	25.0	21.0	17.8		
		MCDB	17.0	18.2	18.6	21.3	23.5	26.9	30.5	30.9	29.1	26.6	22.8	18.9		
	2%	WB	13.4	15.3	16.4	19.0	21.4	24.8	27.1	27.2	26.0	23.3	19.4	15.1		
		MCDB	14.9	16.8	17.9	20.6	22.4	26.3	29.6	30.3	28.5	25.5	21.3	16.8		
	5%	WB	11.2	13.1	15.0	17.9	20.4	24.2	26.7	26.8	25.4	21.8	18.2	13.4		
		MCDB	13.0	14.8	17.1	19.6	22.1	25.7	29.1	29.8	28.0	24.1	20.3	15.5		
	10%	WB	9.3	10.9	13.6	16.8	19.5	23.5	26.2	26.5	24.7	20.4	16.8	11.8		
		MCDB	11.7	13.1	15.9	18.8	21.9	25.0	28.7	29.5	27.3	23.1	19.3	14.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.6	7.4	7.7	7.6	5.8	5.1	5.9	6.2	6.9	7.1	6.4		
		MCDBR	9.0	9.0	9.1	9.0	9.4	6.9	6.3	6.8	6.9	7.8	8.0	8.6		
	5% WB	MCWBR	7.0	7.5	7.0	5.8	5.0	3.5	2.7	2.7	3.3	4.4	5.6	6.2		
		MCWBR	8.2	8.2	8.1	6.4	5.9	4.0	5.3	5.9	5.9	6.1	6.8	7.7		
(40) Clear-Sky Solar Irradiance	taub		0.455	0.489	0.544	0.559	0.564	0.577	0.555	0.524	0.501	0.468	0.442	0.442		
		taud	2.007	1.953	1.830	1.849	1.897	1.960	2.059	2.162	2.163	2.139	2.131	2.046		
	Ebn at Noon		727	747	740	751	752	739	753	772	771	762	739	713		
		Edh at Noon	142	166	201	206	198	185	167	149	143	136	125	130		
(44) All-Sky Solar Radiation	RadAvg		2.01	2.83	4.00	4.83	5.14	4.41	5.38	5.50	4.36	3.65	2.52	1.87		
	RadStd		0.22	0.34	0.29	0.41	0.62	0.65	0.72	0.47	0.41	0.34	0.27	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.42	+0.56	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page